Chapter 1

Project Design Document

SDR-Based Spectrum Monitoring System for FM Broadcast Screening and Compliance Assessment Support Continuous Monitoring of FM Broadcast Using HackRF One

Abstract. The document defines the end-to-end design of a low-cost, continuously operating SDR spectrum-sensing system for calibrated screening of FM broadcast services and technical support to compliance assessment. Its current required measurement outputs are per-channel calibrated received power and occupancy; duty cycle, noise-floor estimates, timing information, and related quality indicators are supporting observables. Carrier-frequency error, field strength, occupied bandwidth, adjacent-channel/out-of-band emission, and modulation-related quantities are not current required measurands unless later admitted through a controlled Requirements Baseline revision. The system integrates a HackRF One receiver with signal processing, measurement storage, visualization, and monitoring services. Project development follows the controlled engineering lifecycle defined in `zz_Workflow`: project definition, requirements baselining, architecture and design, build and verification specification, prototype construction, and measured verification. Within that lifecycle, the prototype implementation sequence progresses through hardware characterization and calibration, sensor construction, PSD estimation, channel-power integration, adaptive occupancy detection, database logging, dashboard visualization, field validation, and multi-sensor deployment. The design emphasizes fixed-gain calibrated sensing, quantitative detection and accuracy targets, reproducible validation, operational health monitoring, scalability, and mitigation of practical risks including overload, frequency drift, noise-floor errors, USB failures, and storage growth.

1 Introduction

Lifecycle Role of This Document

This document is a consolidated engineering source document for the SDR prototype project. It contains project-definition context, requirement material, architecture and design decisions, implementation planning, and verification material used by the staged lifecycle defined in `zz_Workflow`. It does not by itself replace or approve the controlled lifecycle artifacts. Where applicable, the authoritative Project Definition Baseline, Requirements Baseline, Architecture and Design Baseline, and Build and Verification Specification are the approved outputs of Ctx 00–Ctx 03, respectively. The numbered phases

in this document describe the prototype implementation sequence and shall not be interpreted as substitutes for those lifecycle stages or readiness gates.

Definition ownership. The acquisition capture plan, estimators, occupancy detector, calibration model, and release rule of the two admitted outputs are defined in `00_sdr_sensor_design`; their validity framework is the Admitted-Output Validity Matrix of `zV`. Where any other document of the set restates either of them differently, those owners prevail and the difference is a defect to be corrected. These definitions are subject to approval in the corresponding lifecycle baselines.

Glossary and Document Set

Each term below has one spelling and one definition site in the document set. An alias in brackets is the only other permitted spelling. A reference to a document of the set means the revision listed in the second table; a change of revision is recorded there.

Objective

Design, build, and field-validate a continuously operating spectrum-sensing sensor that monitors FM broadcast services and records per-channel received power and occupancy over time. The system must be low-cost, reproducible, and scalable from one sensor to a network of geographically distributed sensors reporting to a central database.

Intended User

The intended user is the Agencia Nacional del Espectro (ANE) of Colombia, with operational use by personnel responsible for FM broadcast spectrum monitoring and technical support to compliance assessment.

Operational Environment

The system is intended for continuous field deployment at FM broadcast monitoring sites in Colombia. The operational context includes an individual sensing node and subsequent extension to geographically distributed sensing nodes whose measurements are reported to a central repository. Monitoring is performed from received RF signals together with site-specific reference information, including applicable FM station assignments and calibration or reference measurements.

Authoritative Technical and Regulatory Sources

The regulatory baseline is the Colombian spectrum and FM sound-broadcasting framework administered by the Agencia Nacional del Espectro (ANE). The authoritative regulatory sources are the Cuadro Nacional de Atribución de Bandas de Frecuencias (CNABF) and the Plan Técnico Nacional de Radiodifusión Sonora en Frecuencia Modulada (FM), established within Resolución ANE 105 de 2020 and the amendments in force on the approval date of the Ctx 00 Project Definition Baseline [1, 2, 3]. Supporting technical sources include ITU-R BS.412, ITU-R SM.1268, and the official HackRF One documentation [4, 5, 8]. IEC 62106 and IEEE 802.22 are supporting references only where their

Table 1.1: Terms of the document set.

Term	Defined in	Meaning
Project Definition Baseline	`zz_Workflow`	Approved output of Ctx 00. The spelling "Problem Baseline" is retired.
Requirements Baseline	`zz_Workflow`	Approved, versioned output of Ctx 01.
Admitted output [required measurement output]	`00`, Table 1.3	A measurement output admitted into the Requirements Baseline as required. At present, calibrated channel power P_c and released channel state s_c only.
Supporting observable	`00`, Table 1.3	A quantity retained only to validate and interpret the admitted outputs.
Calibration correction C_{cal}	`00`, Phase 0 and Phase 2	The correction in dB, a function of frequency and declared gain state, applied to the estimated channel power. The spelling C_{amp} is retired.
Duty cycle D_c	`00`, Table 1.3	Supporting statistic of the released states; O_T and P_{occ} are retired spellings.
Regulatory Source List	`00`, Regulatory Source List	The only list of regulatory sources in the document set; each entry has six fields.
Candidate measurand	`zX`	Any quantity described in the set that is not an admitted output. It is not a requirement unless admitted into the Requirements Baseline.
Measurement validity state	`zV`	The state assigned to one admitted output of a record by the Admitted-Output Validity Matrix; it decides the release of that output.
Admitted-Output Validity Matrix [AOVM, validity matrix]	`zV`	The controlled table (identifier AOVM-2) that assigns the validity state and the consequence of each flag for each admitted output.
Optional future assurance proposal	`zV`, source-role statement	The Compliance-Grade, Conditional Compliance-Grade, MQR and MCC framework with its certification, registry, certificate-validity, renewal, revocation, independent-qualification and formal-decision provisions. Measurement validity states and the Admitted-Output Validity Matrix are not part of it.

Table 1.2: Documents of the set and the revision each reference means.

Document	Short form	Role	Rev.
00_sdr_sensor_design	00	Owner of the definitions of the two admitted outputs	A
zV_AdmittedOutputValidity_SpectrumSensingFM0	zV	Owner of the validity framework of the admitted outputs	A
zX_CandidateFuture_SpectrumSensingFM0	zX	Supporting guidance, candidate procedures, future assurance proposals, restatements of 00; binds nothing	A
zz_Workflow	–	Lifecycle, baselines, change control	H6 D4

respective RDS or spectrum-sensing content is applicable. Every instrument used by the project is recorded in the Regulatory Source List below, which is the only list of regulatory sources in the document set.

Regulatory Source Version Rule

Each authoritative regulatory source used by the project shall be recorded by issuing authority, instrument or resolution identifier, applicable version or amendment state, effective date, the clauses used, the measurand or requirement it governs, and whether those clauses were read from the published text (Regulatory Source List). The approved Ctx 00 Project Definition Baseline freezes the regulatory baseline using the instruments in force on its approval date. A subsequent change to an authoritative regulatory source requires controlled reassessment of the Project Definition Baseline and any affected downstream requirements.

Regulatory Source List

The entries below are the Regulatory Source List required by the Ctx 00 Project Definition Baseline of zz_Workflow. It names every instrument that sets a limit, tolerance, or measurement method for the assessed emissions, or that governs how a measured value is expressed or relied on as evidence, with six fields for each entry. Other documents of the set point to this list and keep no list of their own. IEC 62106, IEEE 802.22 and the HackRF One documentation are technical references: they set no limit, tolerance, or method for the assessed emissions and do not govern how a value is relied on, so they are not entries. The list is a draft until the editions in force on the approval date are recorded and a person has compared the clauses with the published text; the last field of each entry states what has been read.

RSL-1

Issuing body	Agencia Nacional del Espectro (ANE), Colombia
Identifier	Anexo 2, Plan Técnico Nacional de Radiodifusión Sonora en Frecuencia Modulada (FM), of Resolución ANE 105 de 2020, as adopted by Resolución ANE 463 de 2020 and amended [2, 9]
Edition or date applied	The text in force on the approval date of the Ctx 00 Project Definition Baseline, to be recorded then. Amendments consulted for this draft: Resoluciones 147/2022, 420/2022, 742/2023, 111/2026, 377/2026, 406/2026 and 581/2026.
Clauses used	5.1.1 (frequency tolerance); 5.1.2 and 3.42 (maximum deviation); 5.1.5 and 5.2.6 (necessary bandwidth, emission class); 5.1.4.1 and 5.1.4.2 (out-of-band and spurious emissions); 3.31 and 5.1.8.1 (minimum usable field strength, ERP classes).
Measurand or requirement governed	Carrier frequency and error, frequency deviation and occupied bandwidth (excluded); field strength at the measurement point and out-of-band and spurious emissions (outside the system boundary); records in Table 1.4. The ERP classes govern the transmitter, not received power, and are not a measurand of this system.
Read from published text	Partly: numerals up to 6.5.5 of the text adopted by Resolución 463 de 2020, and the amendments listed, were read through an extraction tool; numerals 7 to 12 and the Apéndices were not read; not yet compared by a person with the Annex as published by ANE.

RSL-2

Issuing body	ANE, Colombia
Identifier	Apéndice A (channel assignments) of the Anexo 2 of RSL-1, as modified
Edition or date applied	In force on the approval date, to be recorded then. Resoluciones 147/2022, 111/2026, 377/2026 and 581/2026 modify channel data only.
Clauses used	Per-channel assignment record (frequency, class, power, service area, channel state).
Measurand or requirement governed	Channel occupancy and emission presence: s_c is compared with the assignment of the channel (Phase 4 alerts; the site-specific station-assignment reference of this document records the assignments of the monitored site). The assignment list is not a limit, tolerance, or method; it is listed by a recorded choice.
Read from published text	No: the Apéndices were not on the pages read.

RSL-3

Issuing body	ANE, Colombia
Identifier	Resolución No. 105 de 2020 (body), art. 1.5.1, as amended [1]
Edition or date applied	As amended and in force on the approval date, to be recorded then.
Clauses used	1.5.1 (the national technical plans are part of the CNABF).
Measurand or requirement governed	None: frames RSL-1, RSL-2 and RSL-8 and sets no limit.
Read from published text	Partly.

RSL-4

Issuing body	ANE, Colombia
Identifier	Resolución No. 773 de 2023 (exposure to electromagnetic fields) [10]
Edition or date applied	31 October 2023.
Clauses used	Numeral 2 (reference levels, Table 1); numeral 3.4 (measurement method).
Measurand or requirement governed	Field strength or exposure at the measurement point, if the resolution applies to FM broadcast (not confirmed): outside the system boundary (Table 1.4).
Read from published text	Partly: Table 1 was not readable; applicability to FM broadcast is not confirmed.

RSL-5

Issuing body	International Telecommunication Union, Radiocommunication Sector (ITU-R)
Identifier	Recommendation SM.1268-5, method of measuring the maximum frequency deviation of FM broadcast emissions at monitoring stations [5]
Edition or date applied	21 August 2019.
Clauses used	The whole recommendation (a method).
Measurand or requirement governed	Measurement method for frequency deviation (excluded; Table 1.4).
Read from published text	Title and status only.

RSL-6

Issuing body	ITU-R
Identifier	Recommendation BS.412-9, planning standards for terrestrial FM sound broadcasting at VHF [4]
Edition or date applied	December 1998.
Clauses used	None identified.
Measurand or requirement governed	None: planning reference; no limit of this recommendation is used.
Read from published text	Title and status only.

RSL-7

Issuing body	United States Federal Communications Commission (FCC)
Identifier	47 CFR §73.1545(b), carrier frequency departure tolerances [11]
Edition or date applied	Current edition of the eCFR, date to be recorded.
Clauses used	Paragraph (b)(1).
Measurand or requirement governed	None in Colombia: comparative reference for the carrier-frequency tolerance only.
Read from published text	Paragraph (b) read.

RSL-8

Issuing body	ANE, Colombia
Identifier	Cuadro Nacional de Atribución de Bandas de Frecuencias (CNABF) [3]
Edition or date applied	Edition applicable to 87.5–108 MHz not retrieved.
Clauses used	Not retrieved.
Measurand or requirement governed	Requirement FRQ-001 (frequency range 87.5–108 MHz).
Read from published text	No.

RSL-9

Issuing body	ANE, Colombia
Identifier	Manual de Gestión Nacional del Espectro Radioeléctrico, Título V, Monitoreo del Espectro Radioeléctrico [12]
Edition or date applied	Edition not recorded.
Clauses used	Sec. 3.3.3.4.5, Configuración del umbral.
Measurand or requirement governed	Requirement OCC-002 (the 10 dB margin, selected within the 8 dB to 12 dB range of the Manual by analogy; not an FM limit).
Read from published text	No: cited as quoted in the Released-channel-state paragraph; not read in this review.

RSL-10

Issuing body	ISO and IEC
Identifier	ISO/IEC 17025, general requirements for the competence of testing and calibration laboratories [13]
Edition or date applied	Edition not recorded.
Clauses used	None used yet; supporting reference for traceability and uncertainty methods. Accreditation is not required.
Measurand or requirement governed	Requirement PWR-001, as a supporting reference for the traceability of the calibration reference source.
Read from published text	No.

RSL-11	
Issuing body	Joint Committee for Guides in Metrology (JCGM)
Identifier	JCGM 100:2008, Evaluation of measurement data — Guide to the expression of uncertainty in measurement [14]
Edition or date applied	2008.
Clauses used	Clause 6 (expanded uncertainty and coverage factor).
Measurand or requirement governed	Requirement PWR-001 and the OCC-001 reporting: how the expanded uncertainty U_c and its coverage factor are expressed.
Read from published text	No: the clause number is not from a reading in this review.

System Overview

The sensor chain is:

$$\textit{Antenna} \rightarrow \textit{HackRF One} \rightarrow \textit{GNU Radio DSP} \rightarrow \textit{measurement writer} \rightarrow \textit{time-series DB} \rightarrow \textit{web dashboard}$$

Figure 1.1 shows the end-to-end architecture for both the single-sensor (Phases 0–5) and multi-sensor (Phase 6) configurations.

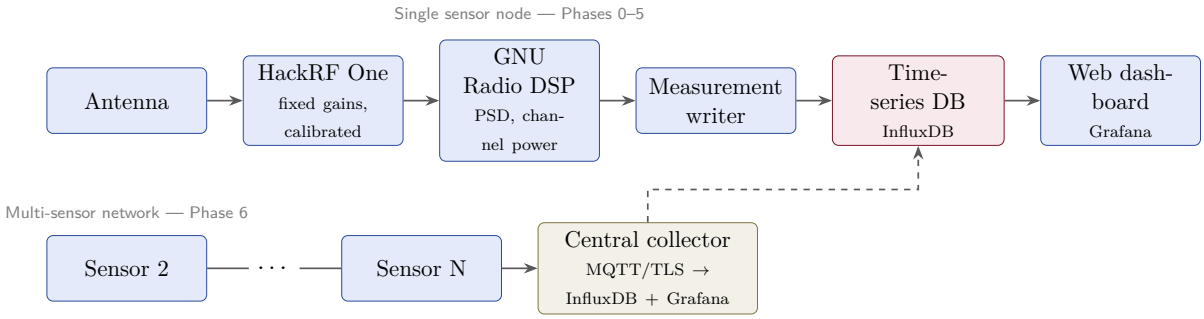


Figure 1.1: End-to-end architecture. Top: single-sensor configuration (Phases 0–5). Bottom: multi-sensor extension (Phase 6); distributed sensors publish over MQTT/TLS to a central collector that feeds the shared time-series database and dashboards.

Scope

Authoritative Project Boundary — Current Scope. The project is limited to calibrated screening and technical support to compliance assessment. The system may provide calibrated measurements, monitoring records, quality indicators, uncertainty information, alerts, and other technical evidence to support review by the responsible regulatory personnel. It shall not independently establish formal regulatory compliance or non-compliance.

The complete Compliance-Grade / Conditional Compliance-Grade / MQR / MCC qualification framework is not required by the current project scope and shall not be incorporated as a system requirement. References to that framework in supporting technical sources shall be treated as design or assurance proposals outside the current Requirements Baseline unless a future documented stakeholder-approved scope change explicitly introduces formal compliance-grade measurement certification.

Calibration, traceability, uncertainty or quality characterization, acquisition integrity, measurement-validity checks, hardware limitations, and reviewable technical evidence remain applicable where source-supported and necessary for screening/support. This boundary does not establish any new operational facts or approve additional measurands.

Requirements-Admission Rule. Where supporting technical sources specify differing acquisition configurations, gain strategies, calibration procedures, hardware-capability claims, or additional measurands, those statements remain non-normative candidate evidence until explicitly admitted into the versioned Requirements Baseline. The current baseline shall define only the minimum controls necessary for FM channel occupancy and calibrated received-power screening and shall not inherit additional measurands, certification mechanisms, or implementation-specific controls solely because they appear in a supporting source.

- **In scope:** sensing and logging of FM broadcast occupancy and power; calibration against references; web visualization; multi-sensor aggregation.
- **Out of scope (v1):** demodulation/recording of audio content, transmitter geolocation, real-time regulatory enforcement actions.

2 Phase 0 — Requirements, Measurand Definition, and Calibration

Measurand Definition

The current required measurement outputs are defined per FM channel c with centre frequency f_c and channel bandwidth B_c. Supporting observables are retained only to validate and interpret those outputs: The duty cycle is evaluated over a declared reporting interval of full-band scans, each contributing one released state s_c per channel. With $N_{\mathrm{Occ}}(c)$, $N_{\mathrm{Free}}(c)$ and $N_{\mathrm{Unc}}(c)$ the numbers of scans of the interval in which s_c is Occupied, Free and Uncertain,

$$D_c = \frac{N_{\mathrm{Occ}}(c)}{N_{\mathrm{Occ}}(c) + N_{\mathrm{Free}}(c)}, \qquad U_c = \frac{N_{\mathrm{Unc}}(c)}{N_{\mathrm{Occ}}(c) + N_{\mathrm{Free}}(c) + N_{\mathrm{Unc}}(c)}, \tag{1.1}$$

and U_c, the fraction of Uncertain scans, shall be reported with D_c. Uncertain scans are counted as neither Occupied nor Free. The record retains the three counts, the reporting interval, and the identifiers of the detector configuration and of the validity-matrix revision in force. D_c and U_c are statistics of the released states and do not replace s_c. When $N_{\mathrm{Occ}}(c) + N_{\mathrm{Free}}(c) = 0$ in an interval, D_c is not defined and is not reported for it, and the alert rule that reads the duty cycle has no input from that interval. This is the only definition of the occupancy statistic in the document set: the names O_T and P_{occ} used in `zX` are retired and denote D_c.

Table 1.4 records, for each measurand class considered, whether it is adopted, excluded, or outside the system boundary, and the Regulatory Source List entry on which the record rests.

Table 1.3: Current measurement outputs and supporting observables.

Quantity	**Symbol**	**Unit**	**Role / definition**
Calibrated channel power	P_c	dBm	Required output; integrated power over $B_c = 200$ kHz, by the estimator of Phase 2 (Channel power integration)
Released channel state	s_c	enumerated	Required output; Occupied / Free / Uncertain per OCC-002
Detector decision	o_c	boolean	Supporting; documented detection rule applied to P_c
Noise floor	$P_{nf,c}$	dBm in B_c	Supporting observable for threshold, release, and quality assessment
Duty cycle	D_c	%	Supporting statistic over a reporting interval, defined below; it is not an output
Full-band scan timestamp	t	UTC	Supporting acquisition metadata; start of scan, ms resolution
Validity states of the channel	q_c	enumerated pair	Supporting quality metadata: the pair (`validity_state_occ`, `validity_state_pwr`) of the Admitted-Output Validity Matrix of `zV`, stored with `validity_flags`; the release of s_c and P_c is decided from the pair

Output Package. Each required output is delivered as a package: the value, its uncertainty, the limit that applies or the recorded absence of one, the comparison with that limit, and the record fields that carry them. Table 1.5 traces each element to the section that produces it and the field that stores it. The package is defined only here; `zV` assigns the validity states it uses, and `zX` holds candidate and future guidance for other measurands. *Uncertainty of P_c.* $U_{P,c} = U_{\text{cal}} + r_c$. U_{cal} is the half-width verified for P_c under PWR-001 (Phase 2, Power accuracy); until it is verified, or when u_{ver} of the verification source is not below the verified value (Power Calibration), the provisional 3 dB design target stands in for it, and `power_unc_basis` records `TARGET` instead of `VERIFIED`. The image term is $r_c = 10\log_{10}\left(1 + 10^{(P_{\text{img},c} - P_c)/10}\right)$ when `IMAGE_SUSPECT` is active or the image test is not evaluated, except when `CAL_INVALID` is active, with $P_{\text{img},c}$ the expected image residual of the capture-plan rules, and $r_c = 0$ otherwise. The linear sum is a conservative provisional choice. A `QUALIFIED` power value proceeds to the comparison with r_c in $U_{P,c}$; other qualifying flags are carried in `validity_flags`. The uncertainty of s_c is carried by the state Uncertain and by the statistic U_c defined above.

Limit. A limit applies to P_c only when it is stated in the quantity of P_c: received channel power in B_c, in dBm at the receiver reference plane, with a direction. A limit stated in another quantity (ERP, EIRP, field strength) applies to that quantity and not to P_c (Table 1.4). The limit is a versioned configuration parameter that carries the identifier of its source; no value is built into the design. No listed source sets one, so the three limit fields are absent and the comparison is `NOT_CONFIGURED`. If the closure of DI-10 finds no source, the output remains a value with no limit by a recorded reviewer decision.

Comparison of P_c. The state is assigned in this order. (1) If the power validity state is `INVALID` or `NOT_EVALUABLE`, the state is `ASSESSMENT_UNAVAILABLE` and the value is not released. (2) If no limit is configured, the state is `NOT_CONFIGURED`. (3) Otherwise,

Table 1.4: Measurand classes and their records.

Class	**Record**	**Entry and clauses**	**Reason**
Carrier frequency and frequency error	Excluded	RSL-1, 5.1.1 (± 2 kHz)	Not adopted: the current scope is calibrated screening of occupancy and received power, and the baseline has no frequency reference (Frequency-reference characterization).
Frequency deviation	Excluded	RSL-1, 5.1.2 and 3.42 (± 75 kHz); RSL-5 as method	Not adopted: outside the current scope.
Occupied bandwidth	Excluded	RSL-1, 5.1.5 and 5.2.6	Not adopted: outside the current scope.
Field strength at the measurement point	Outside the system boundary	RSL-1, 3.31 and 5.1.8.1 (66 dBμV/m); RSL-4, numerals 2 and 3.4, if it applies to FM	The baseline has no antenna-factor characterization or field-strength calibration (Candidate future field-strength estimation); RSL-4 prescribes a calibrated RMS probe and a multi-point layout.
Out-of-band and spurious emissions	Outside the system boundary	RSL-1, 5.1.4.1 (25 dB, 120–240 kHz) and 5.1.4.2 (35 dB, 240–600 kHz; $43 + 10 \log P$ beyond)	The far limit exceeds the dynamic range the baseline receiver can demonstrate (about 48 dB for the 8-bit converter); the near limits are not adopted and have not been assessed against the receiver.
Channel occupancy and emission presence	Adopted: output s_c	RSL-2 (assignment record)	s_c is compared with the assignment of the channel; see RSL-2 for the basis of the entry.
Calibrated received power (not a class of the engine list)	Admitted output; no governing entry	None found in the list	No listed entry sets a limit, tolerance, or method for received channel power at the monitoring point, so P_c is not a compliance-relevant measurand. The source, value, and direction of any threshold for it is item DI-10 of the deferred-item register; the comparison state is `NOT_CONFIGURED` until it closes (Output Package).

Table 1.5: Output package: elements, producers, and fields.

Element	Definition	Produced in	Field
Power value	P_c in dBm at the receiver reference plane	Phase 2, Channel power integration	`power_dbm`
Power uncertainty	$U_{P,c}$, half-width in dB, with its basis	This paragraph	`power_unc_db`, `power_unc_basis`
Power limit	T in dBm at the same plane, direction `UPPER` or `LOWER`, source identifier; absent while no limit is configured	This paragraph; configuration	`limit_dbm`, `limit_dir`, `limit_source`
Power comparison	One of five states	This paragraph	`power_comparison`
Occupancy output	s_c	Phase 2, Released channel state	`state`
Assignment	Entry of the site-specific station-assignment reference for the channel	Station-assignment reference	`assigned` (bool)
Occupancy comparison	One of four states	This paragraph	`occupancy_comparison`
Validity states	Occupancy and power	`zV`	`validity_state_occ`, `validity_state_pwr`

with P_c and T in dBm and $U_{P,c}$ in dB, for an `UPPER` limit

$$\text{state} = \begin{cases} \texttt{WITHIN_LIMIT}, & P_c + U_{P,c} \le T, \\ \texttt{OUTSIDE_LIMIT}, & P_c - U_{P,c} > T, \\ \texttt{INDETERMINATE}, & \text{otherwise}, \end{cases} \tag{1.2}$$

and for a `LOWER` limit `WITHIN_LIMIT` when $P_c - U_{P,c} \ge T$, `OUTSIDE_LIMIT` when $P_c + U_{P,c} < T$, and `INDETERMINATE` otherwise. The states `INDETERMINATE`, `NOT_CONFIGURED` and `ASSESSMENT_UNAVAILABLE` are never converted to `WITHIN_LIMIT` or `OUTSIDE_LIMIT` by averaging, rounding, a small interval, or a default, and `NOT_CONFIGURED` is never displayed or counted as `WITHIN_LIMIT`; a state changes only through the result of a later scan. These states are screening observations for personnel review and are not compliance determinations. `INDETERMINATE` is distinct from the occupancy state Uncertain and from the formal decision states of the optional future framework in `zX`.

Comparison of s_c. Each scan compares the released state with the assignment reference:

s_c	Assigned	`occupancy_comparison`
Occupied	yes	`CONSISTENT`
Occupied	no	`UNASSIGNED_OCCUPIED`
Free	yes	`ASSIGNED_FREE`
Free	no	`CONSISTENT`
Uncertain	either	`INDETERMINATE`

An Uncertain state is never counted as `CONSISTENT` or `ASSIGNED_FREE`.

Screening and alert rules. The screening rules of this system are the two comparisons above. Its alert rules are those of Phase 4 (Alerting Rules), which read the fields of this package. No other screening or alert rule is adopted for the admitted outputs; a rule that compares P_c with a limit is added only when DI-10 closes.

Table 1.6 summarizes the top-level requirement material carried by this design document. These targets characterize screening/support performance; meeting them does not certify regulatory compliance. Their normative status is governed by the approved Requirements Baseline and its controlling Project Definition Baseline; this document does not independently admit, approve, or revise a requirement. Requirement identifiers shall match those of the approved Requirements Baseline. They are distinct from the R0–R3 change-control connectors of `zz_Workflow`, and the bare form Rn shall not be used for requirements. Verification procedures and open-item status are mapped in Table 1.9.

Table 1.6: Top-level requirements.

ID	Requirement	Target
FRQ-001	Frequency range	87.5–108 MHz
SCN-001	Full-band scan period	≤ 10 s
OCC-001	Occupancy detection	$P_{fa} \leq 5\%$, $P_d \geq 95\%$ at SNR ≥ 0 dB
OCC-002	Free-state release guardrail	Sensitivity margin ≥ 10 dB (Section 4)
PWR-001	Absolute power accuracy	± 3 dB (provisional design target, post-calibration)
MAP-001	Channel-frequency mapping integrity	± 5 kHz at 100 MHz (supporting acquisition criterion
AVL-001	Uptime	$\geq 95\%$ over a 30-day run
SCL-001	Scalability	≥ 5 sensors feeding one central DB

HackRF One Hardware Characterization

Gain chain: The HackRF receiver path provides three digitally controlled gain stages: RF amplifier (AMP, 0/14 dB, fixed step), LNA (0–40 dB, 8 dB steps), and VGA (0–62 dB, 2 dB steps). The current calibrated-screening baseline uses declared **fixed** gains. Individual gain settings and amplifier state shall be recorded with each acquisition, and clipping/integrity checks remain mandatory because a fixed gain value alone does not demonstrate valid acquisition. AGC is not part of the current baseline; it may be evaluated as an alternative only after separate validation and explicit admission into the Requirements Baseline. Recommended starting point for a moderate-signal urban/rooftop site: RF amp off (0 dB), LNA = 16 dB, VGA = 20 dB, then adjust per site using the overload checks in Section 10.

Noise figure and dynamic range: The noise figure is estimated in bench tests using the *gain method*:

$$NF \approx P_{n,\text{out}} \text{ (integrated in } B) \; - \; (-174 + 10\log_{10} B) \; - \; G_{\text{total}} \quad \text{[dB]}, \tag{1.3}$$

where $P_{n,\text{out}}$ is the measured output noise power with the input terminated in 50 Ω, B the measurement bandwidth, and G_{total} the total applied gain. The usable dynamic range and compression behavior shall be characterized empirically for the

deployed gain configuration. No universal 60 dB dynamic-range requirement is imposed by the current baseline; adequacy is established by validation of the admitted occupancy and calibrated-power outputs under the intended signal environment.

Frequency-reference characterization. The stock crystal is specified at roughly ±20 ppm; at 100 MHz this is up to ±2 kHz. For the current occupancy/power baseline, the frequency-reference state and any correction needed to preserve channel-frequency mapping shall be recorded per device. Carrier-frequency error is not a current required measurand. A TCXO, GPSDO, or equivalent disciplined reference becomes mandatory only if a later admitted measurement objective requires accuracy or stability that cannot be demonstrated with the baseline reference/verification method.

Power Calibration: HackRF reports uncalibrated digital power; calibrated received-power screening therefore requires an amplitude-correction model tied to a defined receiver reference plane and to the active fixed-gain configuration. The correction is written $C_{\mathrm{cal}}(f, g)$, in dB, where g denotes the declared gain state (gain stages and amplifier state); this is the only name used for it in the document set. It is applied in the estimator of Phase 2 (Channel power integration), which is the single definition of P_c. The correction shall be established at enough frequencies and gain conditions to demonstrate the required power accuracy over the operating band; using at least five frequencies is a reference implementation, not a universal requirement. The calibration record identifier, applicable hardware/firmware configuration, reference plane, reference source, and validity state shall be retained. The correction shall be established against a signal source or power meter whose level at the receiver reference plane is traceable to a national or international standard, and its identifier, calibration date, and uncertainty are recorded in the calibration record. This calibration reference shall not be the T3 reference used to verify PWR-001 (Phase 2): the two shall be different instruments with independent traceability, and PWR-001 is not verified when they coincide. The identity of the traceable source is item DI-09 of the deferred-item register. Periodic verification compares the calibrated power P_c (by the estimator of Phase 2) of a continuous-wave source of known level at the receiver reference plane, placed at the centre of a channel, with that level. The operator performs it, on the bench or in place, at the active gain state and at intervals not exceeding the declared interval T_{ver} (item DI-09), with at least one verified frequency inside the served span of each capture that uses the record. The capture of a verification is diagnostic: it is made outside the scan plan while the scan is stopped, carries a verification identifier in its capture metadata, writes no admitted-output record, and is exempt from `CAL_INVALID`; a capture without the identifier is never exempt. Every verification, passed or not, is retained in an append-only log of the calibration record with its time, capture, gain state, frequency, source identifier, known level, u_{ver}, U_{cal} in force, r_{ver} and outcome. The verification residual r_{ver}, in dB, is the calibrated power minus the known level. Let U_{cal} be the calibration term of $U_{P,c}$ in force (Uncertainty of P_c) and u_{ver} the uncertainty of the known level at the reference plane in that verification, including the repeatability of the reading; the source may be the calibration source or another. A verification at a frequency counts for every capture whose served span contains it and is logged against each; the verified frequencies of a capture are those declared for it in the record, at least one in a channel that it serves, outside the

DC guard. A verification with $u_{\text{ver}} \geq U_{\text{cal}}$ cannot demonstrate the correction and counts as not performed. The active correction can no longer be demonstrated when $|r_{\text{ver}}| > U_{\text{cal}} - u_{\text{ver}}$ at a verified frequency, or when no verification that passes this criterion has been performed within T_{ver} before the start of the scan, counted from the first passing verification of the record or from the last such verification, separately for each capture and gain state; `CAL_INVALID` is then raised for the captures whose served span contains a failing frequency or, for a lapse, whose interval has expired, and calibrated-power reporting is not evaluable (`zV`). A record that declares no T_{ver}, or a verification that states no u_{ver}, cannot show that the correction is verified, and `CAL_INVALID` is raised. It clears when a valid record is activated after verification or, for a drift or a lapse, when a verification that passes the criterion is performed at every frequency that failed (for a lapse, at the verified frequencies of the capture); the absence of an active valid correction clears only by the activation of a valid record, and the flag clears when no cause remains. A record is activated only at a scan boundary and after its plan check, which rejects the activation if it fails; a candidate record is applied by the verification capture only. A verification interrupts the scan; it is announced by a health point with the state verification, which suppresses the sensor-offline alert, and it counts as downtime for AVL-001. A record applies only to the hardware and firmware configuration stated in it, and any other configuration has no active valid record.

Candidate future field-strength estimation — outside current baseline: If field strength is later admitted as a required measurand, a calibrated antenna factor $AF(f)$ for the monitoring antenna may support:

$$E\,[\text{dB}\mu\text{V/m}] = P_r\,[\text{dBm}] + 107 + AF(f)\,[\text{dB/m}], \tag{1.4}$$

since a 0 dBm signal in a 50 Ω system corresponds to 107 dBμV. $AF(f)$ is taken from the antenna datasheet or measured in an open-area test against a reference field.

3 Phase 1 – Sensor Build (HackRF One + GNU Radio)

Flowgraph. A capture-based flowgraph acquires a validated instantaneous bandwidth per capture; channel selection, filtering, and power integration happen in the DSP stage (Phase 2). The sample rate is a declared deployment parameter and shall be validated for sustained operation on the actual SDR/host configuration, including acceptable sample-loss and host-buffer-overrun performance. Figure 1.2 shows the block topology. Table 1.7 gives the current baseline and a non-normative 20 MS/s coverage example. The number of captures and their centre frequencies shall be derived from the validated usable span while covering 87.5–108 MHz with the required margin. If sustained 20 MS/s operation is demonstrated, the 93/103 MHz two-capture arrangement remains a permitted reference implementation only if it also satisfies the capture-plan rules below; otherwise additional captures may be used.

Capture-plan rules. The admitted outputs are produced from these wideband captures; the offset-tuned single-target acquisition of `zX` is the reference for the candidate

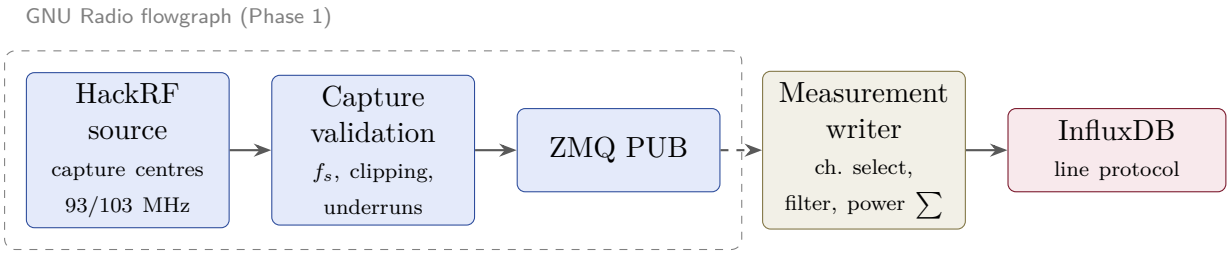


Figure 1.2: Phase 1 capture-based flowgraph block topology. The thin GNU Radio flowgraph (dashed) acquires a validated instantaneous bandwidth per capture and streams IQ over ZMQ; channel selection, filtering, and power integration run in the writer DSP stage (Phase 2).

Table 1.7: HackRF source parameters (Phase 1).

Parameter	**Value**	**Note**
Sample rate	Validated deployment value	20 MS/s is a candidate hardware settin
Capture centres	Derived from validated usable span	93/103 MHz example must also satisfy
Baseband filter	Consistent with validated sample rate/span	Must preserve required FM analysis reg
LNA gain	16 dB starting point	Fixed, site-tunable, explicitly logged
VGA gain	20 dB starting point	Fixed, site-tunable, explicitly logged
RF amp	Off (0 dB) starting point	Enable only after overload/linearity vali

carrier-frequency procedure only. The validated usable span of capture j is the range of offsets $[-a_j, +b_j]$ about $f_{\mathrm{LO},j}$ stated in the active calibration record, and its served span is the corresponding frequency range. Each capture j has a DC guard of declared width W_{DC} centred at its LO frequency $f_{\mathrm{LO},j}$, and an I/Q image map $f \mapsto 2f_{\mathrm{LO},j} - f$. A channel is *served* by capture j only when its allocation $f_c \pm 100\,\mathrm{kHz}$ lies inside the validated usable span and does not intersect the DC guard, and the image of the allocation, $2f_{\mathrm{LO},j} - f_c \pm 100\,\mathrm{kHz}$, also lies inside the validated usable span; serving is a property of the capture plan and does not depend on the signals present. The plan assigns each channel of FRQ-001 to exactly one capture that serves it, the *home capture* of the channel; the channel is evaluated, and enters the vacant set, in that capture only. *Image contamination* of channel c in its home capture j means that the expected image residual $P_{\mathrm{img},c} = P_{c'} - \mathrm{IRR}_j(f_{c'} - f_{\mathrm{LO},j})$, from the spectral content at the image location $f_{c'} = 2f_{\mathrm{LO},j} - f_c$, whether or not it lies in a channel of FRQ-001 (its estimate $P_{c'}$ is the uncorrected band power over B_c centred at $f_{c'}$ in the spectrum of capture j itself, before any validity gating, plus $C_{\mathrm{cal}}(f_c, g)$ of the home channel, so that no correction is extrapolated outside the calibrated band), exceeds $P_{nf,c} - \Delta_{\mathrm{img}}$; it is a condition of one scan, raised as `IMAGE_SUSPECT`, and not a property of the plan. The test is not evaluated, and `IMAGE_SUSPECT` is not raised, when the capture floor is not established ($|\mathcal{V}_j| < V_{\min}$, which includes an empty set, so that $P_{nf,c}$ is undefined) or `CAL_INVALID` is active; the matrix then gives the power value at least the state `QUALIFIED` through the flag that left the floor unestablished (`zV`), and, when the floor is unestablished, the image term r_c of its uncertainty is still evaluated (not when `CAL_INVALID` is active, the power value being `NOT_EVALUABLE` then). IRR_j is the image-rejection ratio after I/Q correction, characterized across the usable span for the declared gain state and stored in the calibration record; between characterized offsets the lower of the two neighbouring values is taken;

before item DI-02 closes the calibration record declares IRR_j, and a capture for which it declares none serves no channel; `zX` supplies no bound. A channel that the plan assigns to a capture that does not serve it for the reason of the span or of the image is a plan failure, checked at the start of each scan (`CONFIG_INVALID`) and at each activation of a calibration record, and the plan is verified against FRQ-001 (i). A span or image failure is of scan scope, because the plan itself is not valid; a guard failure is a channel condition (`DC_GUARD`).

The capture plan shall serve every channel of FRQ-001 in every full-band scan. A channel that the plan assigns to a capture whose DC guard its allocation intersects (a plan that no longer serves it, for example after W_{DC} is revised) raises `DC_GUARD` in that capture until the plan assigns it to a capture that serves it. Because the channel allocations are contiguous, any LO frequency inside the band falls within some allocation, so each channel whose allocation contains or abuts an in-band LO must be served by another capture. The 93/103 MHz example would need each capture to serve the channel at the other LO from an offset of more than 10 MHz, beyond the Nyquist edge of a 20 MS/s capture, so it cannot be met; two captures with LOs at 95 and 101 MHz and a usable span of at least 8 MHz on each side of the LO serve every channel; a third capture or shifted centres are then required, and the resulting scan plan is verified against SCN-001. W_{DC} and Δ_{img} (proposed 6 dB) are items DI-01 and DI-02 of the deferred-item register. Until they close, Δ_{img} has its proposed value and W_{DC} the positive value that the capture-plan record declares, which the plan shall declare. The capture-plan record holds the LO frequencies, the assignment of channels to captures, and the values of W_{DC}, Δ_{img} and $V_{\min}$ in force.

Acquisition and calibration checks. The tests below raise the flags of the Admitted-Output Validity Matrix of `zV`, which holds their names, scopes, classes, clear conditions, and the order in which they are evaluated; this paragraph holds the tests only. For capture j, let $a_{\mathrm{pk},j}$ be the largest value of $|I[n]|/128$ and $|Q[n]|/128$ over the retained block and $H_{\mathrm{ADC},j} = -20\log_{10} a_{\mathrm{pk},j}$ dB, with $H_{\mathrm{ADC},j} = +\infty$ when $a_{\mathrm{pk},j} = 0$; the value of $H_{\mathrm{ADC},j}$ is retained in the capture metadata.

`ADC_CLIP`. Any retained raw sample of the capture equals 127 or -128.

`HEADROOM_LOW`. $H_{\mathrm{ADC},j} < 6$ dB and `ADC_CLIP` is not raised. The 6 dB value and the occupancy class of this flag are a proposal of the matrix and item DI-12 of the deferred-item register.

`GAIN_STATE_MISSING`. The LNA gain, the VGA gain, or the amplifier state of the capture is absent from its record, unknown, or not traceable to it.

`SPECTRUM_INVALID`. The averaged periodogram of the capture holds fewer than M complete FFT frames, or a non-finite PSD value in the allocation of a channel assigned to the capture.

`CONFIG_INVALID`. Checked at the start of each scan: the sample rate, FFT size, window, or averaging count is not that of the active calibration record; AGC is enabled; or the configuration metadata are incomplete, which includes a capture-plan record that declares no positive value of W_{DC}, an active record that states no averaging count or no usable span, and a channel assigned to a capture that does not serve it for the reason of the span or of the image under the active record. With no

active calibration record, the comparison with the record and the plan check are not made and are not counted as failed; the flags that need the record are raised as their triggers state (`CAL_INVALID`, `TEMPLATE_INVALID`), and the other checks of this item are still made.

`CAL_INVALID`. The capture has no active valid amplitude correction $C_{\mathrm{cal}}(f, g)$ for its frequencies, gain state, and reference plane, or periodic verification has found that the active correction can no longer be demonstrated, including the lapse of the verification interval (criterion under Power Calibration).

`TEMPLATE_INVALID`. The capture has no active valid noise-response template $S_j(k)$ for its sample rate, FFT grid, window, and gain state.

The flags `DC_GUARD` and `IMAGE_SUSPECT` are raised by the capture-plan rules above, and `VACANT_SET_INSUFFICIENT` by the detector (Phase 2). `RX_OVERFLOW` and `FRAME_DROPPED` are raised by the writer rules (Section 5). *Limitation.* The tests of this paragraph see the digitized samples only. An overload before the converter, such as compression of the front end by a strong signal inside or outside the band, does not necessarily clip or lower the headroom of the samples, and no flag detects it at run time. The states `VALID` and `QUALIFIED` hold only for input levels within those over which the usable dynamic range was characterized. The overload is bounded by the characterization of the usable dynamic range and compression behavior for the deployed gain configuration (Section 2), by the validation of the admitted outputs under the intended signal environment, and by the overload checks of the risk register (Section 10); it remains a limitation of the admitted outputs.

Bench Bring-up Checklist

1. `hackrf_info` and `hackrf_transfer -r` CLI smoke test (driver and USB integrity).
2. Flowgraph runs 60 s with no receive-overflow (sample-loss) indications reported by the driver stack and no dropped frames detected by the writer.
3. Log, per full-band scan, the receive-overflow count reported by the driver stack and the dropped-frame count detected by the writer (Section 5) for watchdog and validity use (Section 10).

4 Phase 2 — DSP Pipeline and Validation

Signal Processing Chain

PSD estimation. The averaged periodogram over M FFT frames is

$$\widehat{S}_{xx}(k) = \frac{1}{M} \sum_{m=1}^{M} \frac{1}{N_{FFT}\, f_s\, U} \left| \sum_{n=0}^{N_{FFT}-1} w(n)\, x_m(n)\, e^{-j2\pi kn/N_{FFT}} \right|^2, \qquad U = \frac{1}{N_{FFT}} \sum_{n} w^2(n), \tag{1.5}$$

with the declared analysis window $w(n)$ and a validated FFT configuration. The default window is Blackman–Harris, matching the reference implementation in `zX`; the window, N_{FFT}, M, and f_s used for measurement shall be those used to establish

the active calibration record. For $N_{FFT} = 8192$, the bin width is $\Delta f = f_s/N_{FFT}$; the familiar $\approx 2.44\,\mathrm{kHz}$ value applies only when $f_s = 20\,\mathrm{MS/s}$. Likewise, the PSD update duration depends on f_s, N_{FFT}, and M. The selected combination and resulting full-band scan period shall be verified against SCN-001 on the deployed platform rather than inferred from a nominal sample rate.

Channel power integration. Channel power is the band-integrated PSD, not a peak pick. With $\Delta f = f_s/N_{FFT}$, with $\mathcal{K}_c$ the bins of the channel allocation $f_c \pm 100\,\mathrm{kHz}$, and with the correction C_{cal} of Phase 0 applied per bin,

$$P_c\,[\mathrm{dBm}] = 10\log_{10}\left(\sum_{k\in\mathcal{K}_c} \widehat{S}_{xx}(k)\,10^{C_{\mathrm{cal}}(f_k,g)/10}\,\Delta f\right), \tag{1.6}$$

and reduces to a single offset $C_{\mathrm{cal}}(f_c, g)$ added to the uncorrected band power, in dB, when the active calibration record shows that $C_{\mathrm{cal}}(f_k, g)$ varies by no more than a declared bound ϵ_C (proposed 0.1 dB) across $\mathcal{K}_c$. The form used and ϵ_C are recorded with the calibration record; ϵ_C is item DI-15 of the deferred-item register. This is the only formula for P_c in the document set and the single definition of the quantity measured by PWR-001; `zX` points to it and restates none.

Occupancy detection. The detector runs separately on each capture j of a full-band scan, in three steps, and uses no decision from a previous scan.

(1) Per-bin normalization. Let $S_j(k)$ be the receiver noise-response template for the capture's sample rate, FFT grid, window, and gain state, normalized to unit mean over the usable span. It is measured on the bench with the input terminated in $50\,\Omega$, in the same characterization as the noise-figure test, and is part of the active calibration record. The normalized spectrum

$$R(k) = \widehat{S}_{xx}(k)/S_j(k) \tag{1.7}$$

removes the receiver's frequency-dependent noise shape, including passband ripple and filter roll-off, so that vacant channels anywhere in the capture share one noise distribution. The normalized power of channel c is

$$Z_c = 10\log_{10}\Big(\sum_{k\in\mathcal{K}_c} R(k)\,\Delta f\Big). \tag{1.8}$$

(2) Noise reference and per-bin threshold. A provisional floor $Z^{(0)}_{n,j}$ is the median of Z_c over the channels assigned to capture j that are absent from the site-specific station-assignment reference and have $v^{(0)}_c = 1$ (the pre-detector value defined under Released channel state). With $\lambda^{(0)}_j = 10^{Z^{(0)}_{n,j}/10}/B_c$, the per-bin threshold is

$$T_j(k) = \lambda^{(0)}_j\,S_j(k)\,10^{\Delta_{\mathrm{det}}/10}, \tag{1.9}$$

where Δ_{det} is the per-bin margin of this detector (provisional 8 dB; item DI-04 of the deferred-item register). It is defined and owned here, and this subsection is the single statement of its role; the same symbol in a candidate procedure of `zX` names a separate parameter of that procedure. The per-bin threshold has one role only: it excludes from the noise reference any channel that contains a bin with

$\widehat{S}_{xx}(k) > T_j(k)$. It does not attribute a detection to the DC or image artefacts, which the capture-plan rules handle through `DC_GUARD` and `IMAGE_SUSPECT`, and it is not the occupancy decision, because an FM signal at 0 dB channel SNR, spread over B_c, raises individual bins only about 3 dB above the floor on average.

(3) Cross-channel decision. The vacant reference set $\mathcal{V}_j$ contains the channels assigned to capture j that are absent from the assignment reference, have $v_c^{(0)} = 1$, and contain no bin above $T_j(k)$. Then

$$Z_{n,j} = \mathrm{median}_{c\in\mathcal{V}_j} Z_c, \qquad \hat{\sigma}_j = \max\Big(1.4826\ \mathrm{MAD}_{c\in\mathcal{V}_j}(Z_c),\ \sigma_{\min}\Big), \tag{1.10}$$

$$E_c = Z_c - Z_{n,j}, \qquad o_c = \mathbb{1}\{E_c > k\,\hat{\sigma}_j\}. \tag{1.11}$$

$\sigma_{\min}$ is the larger of the theoretical spread of Z_c for noise alone, $\sigma_{\mathrm{th}} \approx (10/\ln 10)/\sqrt{N_{\mathrm{ind}} M}$ dB with N_{ind} the bandwidth B_c divided by the equivalent noise bandwidth of the window and M the number of independent frames averaged, and the spread of Z_c measured on a terminated input at the configured N_{FFT} and M by the estimator 1.4826 MAD used for $\hat{\sigma}_j$ (item DI-03). The measured value includes the finite averaging of the template $S_j(k)$, which σ_{th} omits, and until DI-03 closes $\sigma_{\min}$ is σ_{th}, a lenient bound. It prevents a degenerate MAD from producing false alarms. If $|\mathcal{V}_j| < V_{\min}$, the capture raises `VACANT_SET_INSUFFICIENT` and its channels are not evaluable for occupancy. When $\mathcal{V}_j$ is empty, $Z_{n,j}$ and $P_{nf,c}$ are undefined. The detector runs on capture j only if at least one channel assigned to it has $v_c^{(0)} = 1$; for any other capture neither `VACANT_SET_INSUFFICIENT` nor `IMAGE_SUSPECT` is evaluated, and its channels take their states from the first-pass flags of the matrix. A site supports s_c from this detector only if every capture has at least $V_{\min}$ channels absent from the assignment reference; this is checked at site selection (Phase 5) and recorded, because at a site where nearly all channels are assigned every channel would be not evaluable and s_c would be Uncertain throughout.

With $\lambda_j = 10^{Z_{n,j}/10}/B_c$, the calibrated noise level and the absolute detection threshold of channel c are

$$P_{nf,c}\,[\mathrm{dBm}] = 10\log_{10}\Big(\lambda_j \sum_{k\in\mathcal{K}_c} S_j(k)\, 10^{C_{\mathrm{cal}}(f_k,g)/10}\,\Delta f\Big), \qquad \gamma_c\,[\mathrm{dBm}] = P_{nf,c} + k\,\hat{\sigma}_j. \tag{1.12}$$

The decision o_c depends only on ratios to the capture floor and is therefore independent of the amplitude calibration; $P_{nf,c}$ and γ_c are calibrated and enter the release rule. k is fixed on the tuning set of the Phase 2 validation (Detector performance), with an initial range of 3–4. Whether this detector meets OCC-001 at 0 dB SNR depends on the residual spread $\hat{\sigma}_j$ after normalization and on k; that evidence belongs to the Ctx 02 feasibility basis. $V_{\min}$ (proposed: 10 channels per capture), $\sigma_{\min}$, and the selected k are items DI-03 to DI-05 of the deferred-item register of Section 9. The detector decision o_c is a supporting quantity; the released occupancy output is the channel state s_c defined below.

Released channel state (OCC-002). The required occupancy output is the released

state $s_c \in \{\text{Occupied}, \text{Free}, \text{Uncertain}\}$:

$$s_c = \begin{cases} \text{Occupied}, & o_c = 1 \text{ and } v_c = 1, \\ \text{Free}, & o_c = 0,\ v_c = 1,\ M_c \geq 10\,\text{dB},\ \text{and } \gamma_c \leq P_{\text{req},c}, \\ \text{Uncertain}, & \text{otherwise}, \end{cases} \qquad M_c = P_{\text{req},c} - P_{nf,c}, \tag{1.13}$$

where $v_c = 1$ when the final occupancy validity state of channel c in that scan, assigned by the Admitted-Output Validity Matrix of `zV`, is `VALID` or `QUALIFIED`. The detector uses the pre-detector value $v_c^{(0)}$, which is 1 when the state assigned by the pass-1 flags of that matrix is `VALID` or `QUALIFIED`; the final v_c is not fed back into $\mathcal{V}_j$ or $P_{nf,c}$. Receive overflow, dropped frames, clipping, a missing gain state, an invalid configuration or spectrum, DC-guard overlap, and image contamination make occupancy invalid there; an invalid calibration record, an invalid noise-response template, and an insufficient vacant set make it not evaluable. In every case s_c is Uncertain unless $v_c = 1$. When no matrix is applied or a condition is unresolved, `zV` assigns `NOT_EVALUABLE` and s_c is Uncertain. $P_{nf,c}$ is the locally verified receiver noise level integrated over B_c, in dBm at the receiver reference plane; and $P_{\text{req},c}$ is the declared minimum received power, at the same reference plane, that the sensor is required to detect on channel c. A channel is released as Free only when a signal at the required level would lie at least 10 dB above the verified noise level and above the active detection threshold. Otherwise, the absence of a detection is not evidence of vacancy and the channel is released as Uncertain.

Because $\gamma_c = P_{nf,c} + k\hat{\sigma}_j$ (Occupancy detection), the threshold condition $\gamma_c \leq P_{\text{req},c}$ is equivalent to $k\hat{\sigma}_j \leq M_c$.

$P_{\text{req},c}$ is a site-configuration parameter admitted through the Requirements Baseline and recorded with the site configuration. Its value is not defined by this document and is item DI-07 of the deferred-item register. If $P_{\text{req},c}$ is derived from a field-strength value, the conversion is configuration data and does not admit field strength as a measurand.

The 10 dB margin is selected within the 8 dB to 12 dB noise-margin range defined by ANE in the *Manual de Gestión Nacional del Espectro Radioeléctrico, Título V—Monitoreo del Espectro Radioeléctrico*, Sec. 3.3.3.4.5, *Configuración del umbral* [12] (RSL-9), for threshold configuration when the local noise level is known. That section also identifies FM/BC 88 MHz to 108 MHz as requiring band-specific background-noise treatment. ANE applies the margin to configure a detection threshold; this project applies the same margin as a sensitivity condition for releasing the Free state. The value is therefore a project selection supported by analogy with the ANE range, not a unique ANE-prescribed FM regulatory limit. It remains subject to FM-specific validation and controlled revision, and it does not replace the adaptive occupancy-detection threshold γ_c. It is also distinct from the per-bin margin Δ_{det}, which serves only noise-reference protection. This paragraph is the single statement of the ANE basis for OCC-002; supporting documents reference it rather than paraphrasing the ANE source.

Upgrade path: cyclostationary detection FM broadcasts carry a 19 kHz stereo pilot, producing strong spectral correlation peaks at cyclic frequencies $\alpha = \pm 19$ kHz. An SCD (spectral correlation density) test on candidate channels discriminates true

FM emissions from noise, interference, or intermodulation products — implemented as a Phase 5/6 refinement to reduce false alarms near strong transmitters.

Station identification (ground truth). Optional RDS demodulation (GNU Radio RDS blocks or the `redsea` CLI on narrowband-recorded IQ) extracts Program Identification (PI) codes and station names, giving positive station identification for validation in Phase 5.

Validation Against Known References

Test signals.

- **T1:** Recorded FM IQ replayed through a second SDR or RF playback path (known power, known frequency).
- **T2:** A locally receivable station with published parameters (frequency, ERP, site coordinates).
- **T3:** An independent reference receiver (RTL-SDR or spectrum analyzer) logging the same band. When T3 is used to verify PWR-001, its applicable amplitude calibration or characterization status, reference plane, and measurement uncertainty over the relevant frequencies and signal levels shall be documented and adequate to determine conformity with the ±3 dB PWR-001 target. An uncharacterized receiver may support qualitative comparison but shall not serve as the reference for PWR-001 acceptance.

Detector performance (OCC-001). The atomic OCC-001 validation observation is one annotated channel–scan pair, evaluated on the detector decision o_c; OCC-002 release behaviour is verified separately (Section 9). Use all annotated observations in the declared validation set; do not select or discard observations after detector outcomes are known.

Ground truth. A ground-truth occupied observation is a channel–scan pair carrying a T1 signal, or a station whose presence in the same period is confirmed by a qualified T3 observation; RDS identification may support but not replace that confirmation. A ground-truth vacant observation is a channel without an assignment in the site-specific station-assignment reference whose vacancy in the same period is confirmed by a qualified T3 observation. Absence from the assignment reference alone is not vacant ground truth.

For each ground-truth occupied observation, estimate SNR from the reference observation, over the same 200 kHz channel allocation $\mathcal{K}_c$ used for P_c and never from the sensor's own P_c, so that the SNR that qualifies an observation is independent of the power estimate the detector acts on. Let

$$Q_c = 10^{P_c^{\mathrm{ref}}/10}, \qquad Q_n = \mathrm{median}_{v \in \mathcal{V}_{GT}} \left(10^{P_v^{\mathrm{ref}}/10}\right), \qquad \mathrm{SNR}_c = 10 \log_{10} \left(\frac{Q_c - Q_n}{Q_n} \right), \tag{1.14}$$

where powers are in mW; P_c^{ref} is the power of channel c over that bandwidth measured by the reference path, which is the T3 integrated power or, for a T1 signal, the injected power at the receiver reference plane; and $\mathcal{V}_{GT}$ is the set of ground-truth-vacant reference channels in the same scan, whose P_v^{ref} are the T3 integrated powers over the same 200 kHz bandwidth. If $Q_c \leq Q_n$, treat the observation as SNR < 0 dB. Compute P_d from all ground-truth occupied observations with

SNR $\geq$ 0 dB and compute P_{fa} from all ground-truth-vacant observations in the same declared validation set:

$$P_d = \frac{TP}{TP + FN}, \qquad P_{fa} = \frac{FP}{FP + TN}. \tag{1.15}$$

Observation diversity. Consecutive scans of one channel are strongly correlated and shall not be treated as independent evidence. Each denominator shall: (i) contain at least 20 observations, the minimum count that resolves empirical rates in 5-percentage-point increments; (ii) draw on at least 10 distinct channels, with no single channel contributing more than 20% of the denominator; and (iii) separate observations of the same channel by at least a declared decorrelation interval τ_{dec}. The values in (ii) and τ_{dec} are proposed defaults pending admission into the Requirements Baseline and are item DI-06 of the register of Section 9.

Tuning and acceptance sets. The declared validation observations are divided into a tuning set and an acceptance set before any detector outcome is known; the two sets share no channel–scan pair and no scan, and each satisfies the diversity constraints above. k and Δ_{det} are fixed on the tuning set, and their values, the set identifier, and the date are recorded before the first acceptance observation is acquired. OCC-001 is computed on the acceptance set only. A change of k, Δ_{det} or σ_{min} after the acceptance set has been acquired voids that set, and a new acceptance set is acquired. P_d and P_{fa} are never reported from the tuning set.

Acceptance statistic. Record TP, FN, FP, TN, the two denominator counts, the number of distinct channels per denominator, and the acquisition configuration used for evaluation, together with the two-sided 95% Clopper–Pearson interval for P_d and P_{fa}. OCC-001 is verified only when the point estimates satisfy $P_d \geq 95\%$ and $P_{fa} \leq 5\%$ under this procedure. The reported interval states how strongly the evidence bounds the target; at 20 observations, 19 detections give a lower bound of approximately 75%. Acceptance on a confidence bound rather than a point estimate would be a Requirements Baseline change. The released state s_c of each annotated observation shall also be recorded, so that the Uncertain fraction per ground-truth class is reported.

Power accuracy and channel-mapping integrity

- Channel mapping (MAP-001): in the deployment configuration, apply a continuous-wave tone from a source whose frequency is traceable with expanded uncertainty $U_{\mathrm{ref}} \leq 1$ kHz at the centre frequencies of at least five channels, including the lowest and the highest channel of FRQ-001 and at least one channel assigned to each capture, at a level that gives SNR $\geq$ 20 dB without clipping. For each tone the mapping error ε_f is the frequency of the PSD peak, by three-point parabolic interpolation, minus the source frequency. MAP-001 is satisfied when $|\varepsilon_f| + U_{\mathrm{ref}} \leq 5$ kHz for every tone. The tone count, U_{ref}, and the SNR are proposals (DI-13). This is an acquisition-integrity criterion, not a current carrier-frequency measurand.
- Power: for each of at least 10 distinct channels, compute $\Delta_c = P_c - P_{c,\mathrm{T3}}$ against the qualified T3 reference, per Section 2, and the expanded uncertainty $U_{\Delta,c}$ of that comparison, with its coverage factor stated. PWR-001 is satisfied only when every compared channel meets $|\Delta_c| \leq 3\,\mathrm{dB} - U_{\Delta,c}$. A channel with $|\Delta_c| > 3\,\mathrm{dB} + U_{\Delta,c}$ fails PWR-001; a channel between these limits is

inconclusive, and PWR-001 shall not be declared satisfied while any compared channel is inconclusive. The validation record shall identify the T3 reference and its applicable uncertainty. When PWR-001 is satisfied, U_{cal} of the output package is the largest $|\Delta_c| + U_{\Delta,c}$ over the compared channels; until then the 3 dB target stands in for it. The ±3 dB target is provisional and is item DI-08 of the deferred-item register of Section 9.

Plausibility check via link budget. For station T2, expected received power:

$$P_r = P_{\text{rad}} - \text{FSPL}(d, f) - L_{\text{clutter}}, \qquad \text{FSPL}\,[\text{dB}] = 20\log_{10}(d_{\text{km}}) + 20\log_{10}(f_{\text{MHz}}) + 32.44. \tag{1.16}$$

where P_{rad} is the published radiated-power value normalized to one declared ERP/EIRP reference basis for all comparisons. Agreement within ±10 dB (allowing clutter uncertainty) screens the calibration chain for gross errors such as a wrong gain state, a missing correction, or a unit error. It is a plausibility diagnostic and does not verify PWR-001.

5 Phase 3 — Measurement Database

Technology Choice

InfluxDB 2.x is the default: native time-series storage, downsampling (retention policies), and direct Grafana integration. PostgreSQL/TimescaleDB is an acceptable alternative if relational sensor metadata must be colocated.

Schema

The fields `limit_dbm`, `limit_dir` and `limit_source` are written only when a limit is configured; the other output-package fields are written for every record of a released value or state. Each calibrated `channel_power` record shall retain the gain state used for that measurement and the `calibration_id` of the applicable calibration record. That record shall resolve to the retained hardware/firmware applicability, receiver reference plane, and calibration validity information defined in Section 2. Historical provenance shall therefore be reconstructable directly from persistent data without relying on temporal proximity to a separate `health` record. Each `channel_power` record shall also retain the inputs of its released state (`detector_occupied`, `gamma_dbm`, `noise_floor_dbm`, `release_margin_db`, and `validity_flags`), so that every released Free state can be re-checked against OCC-002 from persistent data. `validity_flags` uses the canonical flag names of the Admitted-Output Validity Matrix of `zV` and the reserved name `UNRESOLVED_CONDITION`, `validity_matrix_rev` identifies the matrix revision that assigned the two validity states (`none` when no matrix was applied), and `capture_id` identifies the home capture of the channel. Every record is retained with its states and flags; the power value of a record whose power validity state is `INVALID` or `NOT_EVALUABLE` is retained and not released. The processing disposition of `zX` is not stored: it is derived, separately for each output, from `validity_state_occ` and `validity_state_pwr` under `validity_matrix_rev` (`zV`, Release and retention), so a record has an occupancy disposition and a power disposition. Every record is screening evidence by the Authoritative Project Boundary, so no per-record screening label is stored, and reports and

Table 1.8: InfluxDB measurements (line-protocol tags/fields).

Measurement	**Tags**	**Fields**
`channel_power`	`sensor_id`, `freq_mhz`, `location`	`power_dbm`, `state` (occupied/free/uncertain), `detector_occupied` (bool), `noise_floor_dbm`, `gamma_dbm`, `release_margin_db`, `gain_lna`, `gain_vga`, `amp_state`, `calibration_id`, `calibration_state`, `validity_flags`, `validity_state_occ`, `validity_state_pwr`, `validity_matrix_rev`, `capture_id`, `power_unc_db`, `power_unc_basis`, `limit_dbm`, `limit_dir`, `limit_source`, `power_comparison`, `assigned`, `occupancy_comparison`, `sw_version`
`spectrum`	`sensor_id`	`psd` (float array / blob sidecar) — optional waterfall history
`health`	`sensor_id`	`rx_overflows`, `frames_dropped`, `records_published`, `gain_lna`, `gain_vga`, `amp_state`, `temp_c`, `uptime_s`

dashboards identify the records as screening evidence. `sw_version` identifies the software and processing-method version that produced the record.

Writer Service

A small Python daemon subscribes to the flowgraph's ZMQ PUB socket, computes per-channel metrics (Phase 2 math lives here, keeping the flowgraph thin), and batch-writes line protocol (batch size 100, flush 1 s). It also emits the `health` measurement every full-band scan and includes a reconnect/backoff on ZMQ or DB failure. The flowgraph shall stamp every published frame with a monotonically increasing sequence number. Because a ZMQ PUB socket discards messages without notice when a subscriber falls behind, the writer shall count sequence gaps as dropped frames. A full-band scan during which a receive overflow or a dropped frame occurs shall carry `RX_OVERFLOW` or `FRAME_DROPPED` in `validity_flags`; its records are retained, are not used in the acceptance statistics, its power values are not released, and its occupancy states are released as Uncertain under OCC-002. The canonical flags are `RX_OVERFLOW` and `FRAME_DROPPED`; their invalidating class is assigned by the Admitted-Output Validity Matrix.

6 Phase 4 — Web Dashboard

Grafana Panels

1. **Occupancy heatmap** (primary view): frequency × time, colour = channel power. This is the key broadcast-monitoring display.
2. **Live spectrum trace**: current PSD with channel markers and RDS-derived station labels.
3. **Per-channel trends**: 24 h power and duty cycle for selected stations.
4. **Sensor table**: online status, last scan age, noise-floor drift, receive-overflow and dropped-frame counts (multi-sensor ready).

The heatmap and the per-channel trends draw released outputs only (`zV`, Release and retention): the power value of a channel whose power validity state is `VALID` or `QUALIFIED`, and the occupancy state s_c. A power value that is retained and not released is not drawn, and the panel shows the validity states and flags of the channel in its place. The live spectrum trace and the sensor table show supporting and health quantities and are labelled as such. The panels identify the records as screening evidence.

Alerting Rules

Alerts identify screening observations for personnel review; they shall not independently establish formal regulatory compliance or non-compliance. The first two rules read the field `occupancy_comparison` of the output package; no alert rule for P_c is adopted while the power limit (DI-10) is open.

- Channel silent: `occupancy_comparison` = `ASSIGNED_FREE` for $T_{\text{on-air}}$ hours on a station previously $> 90\%$ duty cycle. Uncertain observations do not count toward silence.

- Unexpected-channel/anomalous occupancy: `occupancy_comparison` = `UNASSIGNED_OCCUPIED` on monitored FM channels.
- Sensor offline: no `health` point for 3 scan periods.
- Calibration not demonstrated: `CAL_INVALID` active in any capture of a full-band scan, or the interval T_{ver} expiring within a declared margin.
- Noise-floor drift: median of the defined noise floors $P_{nf,c}$ of records not `INVALID`, restarted at a change of the calibration record, shifts by > 6 dB over 24 h (possible gain fault or interference).

7 Phase 5 — Field Validation

Protocol

1. Select a site with a published FM station list and, for every capture of the plan, at least $V_{\min}$ channels absent from the site-specific station-assignment reference; record the count (regulator data: frequency, ERP, transmitter coordinates).
2. Run the sensor continuously for 72 h. This run is an implementation-progress check; AVL-001 is verified only by the 30-day procedure in Section 9.
3. Independently log the band with the reference receiver (T3) for at least 6 h overlapping the deployment.
4. Collect RDS identifications where possible for ground truth.

Field-Validation Diagnostics

The following metrics are field-validation diagnostics. They are not acceptance criteria and are not part of the current Requirements Baseline; admitting any of them as an acceptance criterion requires a controlled Requirements Baseline revision. Their parameters, including the expected-station filter, shall be declared in the validation plan before the run.

- **Detection rate:** licensed stations detected / licensed stations expected at the site (the declared ERP-and-distance filter defines "expected"). Reference value $\geq 90\%$ of expected stations.
- **Ranking consistency:** Spearman's ρ between measured power ranking and predicted ranking ($P_{\text{rad}} - \text{FSPL}$), using the same declared radiated-power reference basis as the link-budget check. Reference value $\rho \geq 0.8$.
- **Stability:** no unrecovered crashes; watchdog restart count logged; disk usage within budget (see Section 8).
- **Error sources documented:** multipath fading (slow power variation of ± 10 dB is normal for a fixed link over a day), intermodulation from strong FM signals (check phantom products at $2f_1 - f_2$ combinations), and front-end overload (add attenuation and re-run if observed).

8 Phase 6 — Multi-Sensor Deployment

Architecture

Sensor nodes forward data to a central collector via MQTT (topic per sensor: `spectrum/<sensor_id>/c` with the telegraf/agent or the writer publishing directly over TLS. The central instance runs InfluxDB and Grafana; dashboards aggregate all sensors using the `sensor_id` tag. In the current scope aggregation is attribution by `sensor_id`; no inter-node calibration model, cross-node consistency analysis, or aggregation rule is part of the baseline (candidate material in `zX`, Section 5).

Time Synchronization

Cross-sensor comparison requires aligned timestamps:

- Minimum: NTP (Chrony), keeping nodes within tens of ms.
- For interference/correlation studies: GPS PPS or a shared GPSDO; note scan start times with sub-ms precision.

Deployment

Each node ships as a `docker-compose` stack: `flowgraph + writer + telegraf + (node exporter)`. Per-node configuration file holds sensor ID, location, validated sample rate/scan plan, calibration table reference, and fixed gains — identical images across sites, configuration via environment. Storage budget. The raw volume of `channel_power` is bounded from the schema of Table 1.8. The numeric fields (8 B each) are `power_dbm`, `noise_floor_dbm`, `gamma_dbm`, `release_margin_db`, `gain_lna`, `gain_vga`, `power_unc_db`, and `limit_dbm`; the boolean fields (1 B) are `detector_occupied` and `assigned`; every other field of the record is a string (declared bound 32 B). A record therefore has $n_{\text{num}} = 8$, $n_{\text{bool}} = 2$, and $n_{\text{str}} = 15$ fields, so $R = 8 \cdot 8 + 2 \cdot 1 + 15 \cdot 32 = 546$ B with the three limit fields written, and 474 B without them. With $N_c = 103$ channels and one full-band scan per $T_s = 5$ s (17 280 scans per day), the bound is $N_c R \cdot 86400/T_s \approx 0.97$ GB per day per sensor (0.84 GB without the limit fields), before InfluxDB compression, which is not assumed. A 7-day raw retention bounds the raw store at about 6.8 GB per sensor, and 1 h downsampling beyond 7 days keeps long-term storage small. The disk growth measured in the 72 h field run replaces this bound (DI-14). Recalculate for site-specific channel count and scan rate, and whenever the schema changes.

9 Requirement Verification Procedures

FRQ-001 (frequency range). Verified by (i) inspection: the declared capture plan places every channel allocation $f_c \pm 100$ kHz in the FRQ-001 range inside a capture that serves it under the capture-plan rules (usable span, DC guard, image) and assigns it to exactly one such capture;

and (ii) test: a T1 signal at the lowest and at the highest channel of the range produces valid `channel_power` records with $o_c = 1$ in every full-band scan of a declared test run.

SCN-001 (scan period). The scan period is measured as the difference between consecutive full-band scan start timestamps t over a declared run in the deployment configuration. SCN-001 is satisfied when every measured interval is ≤ 10 s and every scan carries a record for every channel of FRQ-001; the benchmark on the deployment hardware verifies that this is sustained, and a shortfall is a failed SCN-001, not a result. An interval interrupted by a recorded watchdog restart is excluded from SCN-001 and counted against AVL-001. Acceptance on a percentile instead of every interval would be a Requirements Baseline change.

OCC-002 (Free-state release). Verified by inspection of the stored records of the OCC-001 validation set and of the 72 h field run. Every record with `state` = Free shall show `detector_occupied` = false, an occupancy validity state of `VALID` or `QUALIFIED`, `release_margin_db` ≥ 10, and `gamma_dbm` $\leq P_{\mathrm{req},c}$. A single violating record fails OCC-002. The number of ground-truth-occupied observations released as Free shall be reported.

AVL-001 (uptime). Over a continuous 30-day window, the sensor is available at time t when a complete full-band scan record exists within the preceding three declared scan periods, the same window length as that of the sensor-offline alert. A full-band scan record is complete as defined for SCN-001, with a record for every channel of FRQ-001; the validity states of the records are not a criterion of AVL-001, and the share of records in each state is reported with the result, together with the verification status of the calibration. AVL-001 is satisfied when the available time is at least 95% of the window. The 72 h field run does not verify AVL-001.

SCL-001 (scalability). At least five sensors concurrently deliver `channel_power` and `health` records to one central database over a declared verification period. SCL-001 is satisfied when, for every sensor, the records are attributable by `sensor_id` and the central record count equals the sensor's `records_published` count for the period.

Validity matrix (AOVM-2). Verified by fault injection on the built prototype, in the deployment configuration. For each of the thirteen rows of the matrix, the triggering condition is produced once alone: receive overflow by throttling the host, a dropped frame by pausing the subscriber, clipping by a full-scale signal, low headroom by a signal below full scale that does not clip, a missing gain state by removing it from the record, a spectrum fault by truncating the frame set, an invalid calibration record by deactivating the amplitude correction alone, a drifted correction by offsetting the reading of the verification source by more than $U_{\mathrm{cal}} - u_{\mathrm{ver}}$ and, at the boundary, by just less, a lapsed verification by advancing the system clock used for the scan timestamps beyond T_{ver}, a source with $u_{\mathrm{ver}} \geq U_{\mathrm{cal}}$, the clearing of a drift and of a lapse by a passing verification at every frequency that failed, and of each by a new record, an invalid template by deactivating the template alone, a DC-guard condition by assigning to a capture a channel allocation that intersects its guard, an image condition by placing a tone at the image of a strong channel and a second by placing, above the level at which the expected image residual exceeds $P_{nf,c} - \Delta_{\mathrm{img}}$, a tone outside the band of FRQ-001 at the image of an in-band channel, an insufficient vacant set by declaring all but fewer than V_{min} channels assigned, an invalid configuration by enabling AGC and, separately, by removing W_{DC} from the capture-plan record, a plan failure by assigning a channel

to a capture whose usable span does not hold its image, a capture with no IRR_j, a record without averaging count, no active record, a capture without the verification identifier, a verification at a frequency in two spans, a passing verification at only some of the failing frequencies, and a passing verification offered against the absence of a correction (which must not clear it), and a candidate-pipeline flag by setting it. The fallback is produced as well: with the matrix deactivated, both states are `NOT_EVALUABLE` and `validity_matrix_rev` is `none`; with one condition made unresolvable, the stored states follow the precedence and `validity_flags` contains `UNRESOLVED_CONDITION`. The coupled conditions are produced as well: `ADC_CLIP` with `HEADROOM_LOW` (only `ADC_CLIP` is stored); each capture-scope first-pass flag that blocks the detector for its capture (`ADC_CLIP`, `GAIN_STATE_MISSING`, `SPECTRUM_INVALID`, `CAL_INVALID`, `TEMPLATE_INVALID`), injected into one capture while the other captures run, so that the stored flags of that capture contain neither `VACANT_SET_INSUFFICIENT` nor `IMAGE_SUSPECT`, and the scan-scope flags, which affect every capture; an insufficient vacant set together with a strong channel at the image of another, so that only `VACANT_SET_INSUFFICIENT` is stored, the power value is `QUALIFIED`, and the image term r_c is in $U_{P,c}$; and, for a channel that two captures serve, an invalid calibration record in the capture that does not report it (the stored states of the channel do not change). The matrix is verified only when, for every injection, the stored `validity_flags`, `validity_state_occ`, `validity_state_pwr`, and `validity_matrix_rev` equal those the matrix prescribes, and the released state and power value follow the release rule. The `HEADROOM_LOW` row is closed (DI-12) by a test of equivalence, not by the overlap of two intervals, because two groups can have intersecting intervals while their rates differ by more than the occupancy decision can tolerate (at 20 observations per group, 20 detections of 20 and 14 of 20 give 95% Clopper–Pearson intervals that intersect). The margins δ_H, one for P_d and one for P_{fa}, each the largest degradation that leaves the occupancy decision acceptable for screening, are declared with their justification from the OCC-001 targets in the Build and Verification Specification before the observations are acquired; the test is one-sided, because only a lower P_d or a higher P_{fa} matters. The groups are the observations with headroom under 6 dB and no clipping and those with 6 dB or more, taken from the acceptance observations at the fixed gain state of the deployment: each observation is assigned to a group by the headroom of its capture alone, before any detector outcome is known, no gain is changed for it, each group satisfies the observation diversity constraints, and a site that yields no low-headroom observation without clipping leaves the row open. The minimum count $n_{\min}$ of each of the four denominators is the smallest n for which the 90% interval of the difference of two rates by the method of Newcombe (Wilson score intervals combined), computed for two equal observed rates of one half and n observations in each group, lies inside $(-\delta_H, +\delta_H)$; it fixes the width of the interval and not the probability of passing, and the count required is the larger of $n_{\min}$ and the count of diversity constraint (i). With the required counts, the occupancy class I stands when the adverse limit of that interval (the lower limit for P_d, the upper limit for P_{fa}) lies inside the margin, for P_d and for P_{fa} (two one-sided tests at 5%); when the whole interval of either rate lies beyond the margin on the adverse side, the occupancy class of the row is `NOT_EVALUABLE`; otherwise DI-12 stays open, the proposed class stands, and further observations are taken. The two groups share one scene only in part, so a difference of the scenes between them is reported with

the result; the result of every test on the row is recorded, and a repetition uses a new set. A revision of the 6 dB value voids the observations. A change of the class is a new revision of the matrix, injected again for this row, and the power class Q of the row is not relaxed by this test. The class `NOT_EVALUABLE`, being of capture scope, withholds occupancy for every channel of the capture, and the acceptance statistics of OCC-001 and OCC-002 are then recomputed under the new revision. The test covers the detector decision; the Free release, which uses the absolute floor, is covered by OCC-002.

Table 1.9: Requirement verification map.

ID	**Procedure**	**Open items (deferred-item register)**
FRQ-001	Section 9	DI-01, DI-02
SCN-001	Section 9	–
OCC-001	Phase 2, Detector performance	DI-03 to DI-06
OCC-002	Section 9	DI-07, DI-12
PWR-001	Phase 2, Power accuracy	DI-08 to DI-11, DI-15
MAP-001	Phase 2, Channel mapping	DI-13
AVL-001	Section 9	Closed by the 30-day run (verification activity)
SCL-001	Section 9	Closed by the $\geq$ 5-sensor run (verification activity)
AOVM-2	Section 9, Validity matrix	DI-09, DI-12
Storage budget	Phase 5, Stability	DI-14

Deferred-item register. The register below holds every open item of this document. Under the register rule of `zz_Workflow`, items whose responsible stage is Ctx 02 are closed or rerouted to a later stage before the Ctx 02 to Ctx 03 gate, items whose responsible stage is Ctx 03 are closed before build readiness, and only items whose responsible stage is verification and whose closure condition is a verification activity remain open when the prototype is built. A value in this document marked proposed or provisional has an entry here.

10 Risk Register

11 Prototype Implementation Milestones

Milestone completion criteria are implementation-progress criteria. Prototype acceptance is decided only against the Requirements Baseline identifiers, using the procedures referenced in Table 1.9. **Terminology**

- *FFT frame:* one N_{FFT}-sample block.
- *PSD update:* averaged periodogram over M FFT frames.
- *Capture:* one contiguous IQ block at a single centre frequency.

Table 1.10: Deferred-item register.

ID	Item	Governs	Stage	Closure condition	Status
DI-01	W_{DC}	FRQ-001	Verification	DC-artefact width measured on the built prototype in the bench characterization; value recorded in the capture-plan record	open
DI-02	Δ_{img} (proposed 6 dB)	FRQ-001	Verification	IRR_j measured across the usable span for the declared gain state; value fixed from it and recorded	open
DI-03	$\sigma_{\min}$	OCC-001	Verification	Spread of Z_c measured on a terminated input at the configured N_{FFT} and M; $\sigma_{\min}$ fixed as the larger of the measured and the theoretical spread	open
DI-04	k (initial 3–4) and Δ_{det} (provisional 8 dB)	OCC-001	Verification	Fixed on the tuning set and recorded before the acceptance set is acquired (Detector performance)	open
DI-05	$V_{\min}$ (proposed 10 per capture)	OCC-001	Ctx 02	Admitted by the reviewer into the Requirements Baseline after the site check of Phase 5 on the intended sites	open
DI-06	Diversity values (ii) and τ_{dec}	OCC-001	Ctx 02	Admitted by the reviewer into the Requirements Baseline	open
DI-07	$P_{\mathrm{req},c}$ values	OCC-002	Ctx 02	Declared per channel in the configuration with the source of each value	open
DI-08	± 3 dB target	PWR-001	Ctx 02	Confirmed or revised by the reviewer in the Requirements Baseline	open
DI-09	Identity of the traceable calibration source and the verification interval T_{ver}	PWR-001	Ctx 03	Named in the Build and Verification Specification with its certificate and uncertainty, independent of the T3 reference, and T_{ver} declared with its basis and u_{ver} below the 3 dB target	open
DI-10	Source, value, and direction of the power limit T (D2-OI-1)	Output package	Ctx 02	A document that names the threshold and the quantity it applies to, read from its published text; or a recorded reviewer decision that no limit is set	open
DI-11	Form of $U_{P,c}$	Output package	Ctx 02	Form admitted by the reviewer, or replaced by the combined uncertainty of the calibration record	open
DI-12	6 dB `HEADROOM_LOW` value, its occupancy class, and the margin δ_H	Matrix	Verification	AOVM-2 verification above: equivalence of P_d and of P_{fa} at headroom under 6 dB without clipping and at 6 dB or more, within the margins δ_H declared before the observations	open
DI-13	MAP-001 tone count, U_{ref}, and SNR	MAP-001	Ctx 03	Fixed in the Build and Verification Specification	open
DI-14	Disk allotment against the storage bound	Operation	Verification	Disk growth of the 72 h run measured and compared with the bound of the storage budget	open
DI-15	ϵ_C (proposed 0.1 dB)	PWR-001	Ctx 03	Fixed in the Build and Verification Specification from the constancy of $C_{\mathrm{cal}}(f_k, g)$ measured in the calibration record	open

Table 1.11: Top risks and mitigations.

Risk	**Severity**	**Mitigation**
Absolute power inaccuracy (HackRF uncalibrated)	High	Per-device calibration table (Sec. 2); fixed gains; periodic periodic verification against a known-level source (Power Calibration).
Front-end overload near transmitters	High	Monitor for intermod products at $2f_1 - f_2$; reduce LNA/VGA; add external attenuator; AMP stays off by default.
USB dropouts and sample loss on long runs	Medium	Log receive overflows and dropped frames in `health`; mark affected scans invalid; watchdog (systemd Restart=always); alert on missing health points.
Frequency-reference drift affecting channel mapping (MAP-001)	Low	Characterize per-device offset/state; log drift trend in `health`; use a disciplined reference only when a later admitted objective requires it.
Noise-floor mis-estimation causing P_{fa} violations	Medium	Receiver-response normalization; cross-channel MAD over an assignment-excluded, per-bin-protected vacant reference set; $\sigma_{\min}$ floor; `VACANT_SET_INSUFFICIENT` when the reference is too small.
DC artefact and I/Q image in wideband captures	High	Capture-plan rules (DC guard, image control, home capture); image rejection characterized across the usable span.
Database disk growth	Medium	Retention policies and downsampling (Sec. 8); growth measured in the 72 h run against the bound of the storage budget (DI-14).

Phase	Milestone	Completion criteria
0	Spec & calibration signed off	Occupancy/power outputs approved; amplitude calibration/verification and occupancy-threshold validation defined
1	Sensor online	60 s clean run, zero receive overflows and zero dropped frames, ZMQ streaming
2	DSP validated	OCC-001, PWR-001, MAP-001 verified on T1–T3; OCC-001 uses the declared channel–scan/SNR evaluation contract, diversity constraints, and recorded confusion-matrix counts
3	Logging live	≥ 1 h of queryable DB data
4	Dashboard live	Heatmap + alerts firing in Grafana
5	Field validated	Field-validation diagnostics evaluated against their declared reference values; 72 h continuous run with no unrecovered failures. AVL-001 is verified by the 30-day procedure (Section 9).
6	Network live	≥ 2 sensors aggregated centrally; SCL-001 (≥ 5 sensors) verified by the procedure in Section 9.

- *Full-band scan:* one complete measurement across the FM band (one or more captures, depending on coverage).
- *Scan period:* time between consecutive full-band scans.
- SNR_c*:* channel-integrated SNR over B_c, used to evaluate OCC-001 (Phase 2 validation); obtained from the reference observation, not from the sensor's P_c.
- E_c*:* detector excess of channel c over its capture floor, in dB, compared with $k\hat{\sigma}_j$.
- M_c*:* OCC-002 release margin, $P_{\mathrm{req},c} - P_{nf,c}$.
- SNR_{peak}*:* peak PSD minus regional median floor, a diagnostic of the carrier pipeline in `zX` (flag `LOW_SNR`); it does not enter occupancy detection or release.
- *Served channel:* a channel that lies inside the validated usable span of a capture, with its image also inside it, and does not intersect its DC guard (capture-plan rules); the plan assigns each channel to one such capture, its home capture.

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[8] Great Scott Gadgets, *HackRF One documentation.*

[9] Agencia Nacional del Espectro (ANE), Colombia, *Resolución No. 463 de 2020*, adopting the Anexo 2 of Resolución ANE 105 de 2020, as amended.

[10] Agencia Nacional del Espectro (ANE), Colombia, *Resolución No. 773 de 2023*, exposure of persons to electromagnetic fields.

[11] United States, Code of Federal Regulations, Title 47, §73.1545, *Carrier frequency departure tolerances.*

[12] Agencia Nacional del Espectro (ANE), Colombia, *Manual de Gestión Nacional del Espectro Radioeléctrico*, Título V, *Monitoreo del Espectro Radioeléctrico.*

[13] ISO/IEC 17025, *General requirements for the competence of testing and calibration laboratories.*

[14] Joint Committee for Guides in Metrology, *Evaluation of measurement data — Guide to the expression of uncertainty in measurement*, JCGM 100:2008.